

ECON 636 600: MACROECONOMIC THEORY I

Assignment 3 (due on 10:00am Tuesday, Oct 9)

1. (30 points) In our lecture, we have considered the growth model with cobb-douglas production function and log utility. We have shown that if $v_0(k) \equiv 0$, then

$$T^j v_0(k) = E_j + F_j \log(k),$$

where

$$\begin{aligned} E_j = & \log\left(\frac{A}{1 + \beta F_{j-1}}\right) + \beta \log\left(\frac{A}{1 + \beta F_{j-2}}\right) + \beta^2 \log\left(\frac{A}{1 + \beta F_{j-3}}\right) + \dots \\ & + \beta F_{j-1} \log\left(\frac{A\beta F_{j-1}}{1 + \beta F_{j-1}}\right) + \beta(\beta F_{j-2} \log\left(\frac{A\beta F_{j-2}}{1 + \beta F_{j-2}}\right)) + \dots \end{aligned}$$

Show that

$$\lim_{j \rightarrow \infty} E_j = \frac{\log\left(\frac{A}{1 + \beta F_\infty}\right)}{1 - \beta} + \frac{\beta F_\infty \log\left(\frac{A\beta F_\infty}{1 + \beta F_\infty}\right)}{1 - \beta}.$$

2. (50 points, 2012 Macro qualifier) Consider the following growth model where the planner chooses a sequence of consumption in order to

$$\begin{aligned} \max \quad & \sum_{t=0}^{\infty} \beta^t \ln(c_t), \\ \text{s.t.} \quad & c_t + k_{t+1} \leq A_t(k_t)^\alpha, t \geq 0, \\ & k_0 > 0. \end{aligned}$$

Suppose that the sequence $\{A_t; t \geq 0\}$ follows a periodic pattern, that is, $A_t = 1$ for $t = 0, 2, 4, \dots$ and $A_t = \lambda > 1$ for $t = 1, 3, 5, \dots$

- (a) Set up the Bellman functional equations associated with odd period and even period.
- (b) Solve for the policy functions applicable to odd periods and policy functions applicable to even periods. (Conjecture and verify that the value function takes the form of

$a_e + b_e \log(k)$ in even periods and $a_o + b_o \log(k)$ in odd periods, where a_e, b_e, a_o, b_o are parameters)

(c) Show that the “steady state” of this dynamic model contains a two-period cycle.

3. (20 points) Consider a growth model with foreign aid:

$$\begin{aligned} \max \quad & \sum_{t=0}^{\infty} \beta^t u(c_t), \\ \text{s.t.} \quad & c_t + k_{t+1} \leq F(k_t, 1) + (1 - \delta)k_t + a, \\ & k_0 > 0. \end{aligned}$$

where $a > 0$ denotes the foreign aid. Let $k_{ss}(a)$ and $c_{ss}(a)$ be the capital stock and consumption in the steady state. Determine $\frac{dk_{ss}(a)}{da}$ and $\frac{dc_{ss}(a)}{da}$.